

International AS and A-level

Psychology (9685)

PS02 Biopsychology, Development and Research Methods 1

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL AS PSYCHOLOGY (9685) PS02 JUNE 2024

There were some very good performances within each of the topics in this examination. Psychological knowledge was the strongest skill demonstrated across all sections, with the best performances being on questions assessing AO1.

Students generally did not achieve the same level of success on questions requiring AO3 skills. Those that were successful were able to give clear evaluation points which were elaborated and explained in terms of specific evidence illustrating their point. By contrast, less successful students made lists of simple strengths or weakness statements or used generic evaluation points only. Quality over quantity remains a useful rule of thumb when it comes to these types of questions.

In terms of AO2/application, this skill was generally stronger within the first two sections of the paper where knowledge of psychological concepts was being applied to unfamiliar contexts and scenarios. Application within the research methods section of the paper was not as effective.

There were incidences of misunderstanding or misinterpreting the requirements of questions in this paper. Students need to ensure they understand the target of different command words in terms of what needs to be included within an answer. For example, 'evaluate' as a command word requires AO3 arguments only and AO1 statements are not going to be creditworthy. It is also worth noting that stating the findings of a study is only creditworthy as AO1 until it is presented in the context of supporting or contradicting an idea or concept, at which point it becomes evaluative.

QUESTION 01

Most students demonstrated clear understanding of the function of Wernicke's area as being responsible for understanding language. Successful students included additional detail on specific location or consequences of damage to an individual's speech. Some included a lot more information than required by a 2-marks question.

QUESTION 02

Students clearly have excellent knowledge of the components of the central nervous system, but in the context of 'functions' this information in isolation was not creditworthy. Students who went on to outline the specific functions of the different components using specialist terminology were more successful than those who wrote in broad terms about communication with, or control of 'the rest of the body', that resulted in responses that lacked the detail for higher level credit.

QUESTION 03

Most students showed awareness that motor neurons connect the central nervous system to skeletal muscles or that motor neurons are what allow for movement of the body. However, there were plenty of students who included more information than necessary, giving a full account of the reflex arc.

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A small number of students suggested that motor neurons were able to send information back to the central nervous system or were in some way responsible for the sensation of pain.

Students need to remember that they should refer to the scenario throughout their answer for application questions.

QUESTION 04

Evaluation seems to be a skill that students found more difficult. Within section A this question saw the highest number of students not attempting responses, and the greatest number of those that did attempt it not achieving any marks. The issue for many was missing the focus of the question.

Knowledge of research studies into plasticity was generally very good, but students who outlined a study and its findings did not meet the evaluative requirement of the question. A small number evaluated the concept of plasticity rather than studies into plasticity.

Successful students gave a clear evaluation point and within their elaboration provided explanation of how and why their point was a strength or weakness. Those who focused on the area of research more broadly (as opposed to a specific study) tended to be more successful as they quickly focused on the AO3 requirements of the question rather than providing AO1 description of a specific study first.

QUESTION 05

Well over half of the cohort achieved level 2 responses or above on this question. Almost all students used Sperry's research and the vast majority had at least basic knowledge of these studies and the different tasks that Sperry used when studying split-brain patients. Weaker responses showed a lack of specialist terminology eg referring to eyes as opposed to visual fields.

Again, students need to keep focused on the question asked. In this case, descriptions of the split-brain procedure or why it is required were not relevant in the context of this question, and neither were evaluations of the methodology.

Successful students produced logical step-by-step procedures with specialist terminology, followed by a clear explanation of the findings obtained in reference to both visual fields/hemispheres, with some reference to the nature of hemispheric lateralisation to explain the findings.

QUESTION 06

Students generally demonstrated good knowledge of the structures involved in synaptic transmission, with accurate use of specialist terminology (although there appeared to be some confusion regarding the difference between hormones and neurotransmitters in some cases).

Knowledge of processes within synaptic transmission was not as strong, particularly the post-synaptic processes. A common misconception appeared to be that neurotransmitters enter the post-synaptic neuron in some way, and another that vesicles travel across the synaptic cleft.

Stronger responses included detail on pre- and post-synaptic processes, with detailed explanation of excitation and inhibition and/or summation, with correct use of specialist terminology throughout the answer.

QUESTION 07

There were some very pleasing performances on this question, with about a quarter of students achieving full marks. There is good knowledge of the action of adrenaline, and detailed knowledge on the actions of the sympathetic nervous system. Responses were somewhat less detailed on the actions of the parasympathetic nervous system, but students clearly understand these two branches and their functions in relation to each other.

Good responses applied to the scenario from the outset, clearly outlining the functions of the sympathetic nervous system from before the concert, using terminology and illustrating with the examples from the scenario explaining why the change was necessary/what it allowed to happen. They then repeated this for the parasympathetic activity after the concert was over.

Weaker students did not apply clearly, presenting too much knowledge in isolation, or their application consisted of simply repeating elements of the scenario within their answer with no explanation of this using their own knowledge. Quotes from the scenario often read like after thoughts added to the end of sentences of knowledge, making the application less effective. In addition, there did appear to be a lack of clarity in responses in relation to whether the sympathetic nervous system controls adrenaline release, or if adrenaline controls the sympathetic nervous system. Reference to sickness being connected to immunosuppression was not relevant here as that is a result of chronic rather than acute stress.

QUESTION 08

This question was generally well answered.

QUESTION 09

There were some excellent responses to this question with over half of the students achieving the top level. Most students were able to correctly identify the stage and explain what a child can and cannot do during this. This was done using the specialist terms for the skills, and the tasks used to study child development. Again, many responses included additional detail that was not required eg descriptions of the three mountains task, or procedures for conservation/class inclusion testing.

Students who were not as successful on this question lacked the specialist terminology associated with the topic, or simply listed the different skills without stating what a pre-operational child can or cannot do in relation to these, failing to outline the stage of development as a result.

Students need to be careful about the clarity of the language they use as those who suggested that skills were improving throughout the stage or that these skills were learnt during the stage tended to produce less effective responses as their understanding of where the pre-operational stage ended, and concrete operational stage became blurred.

QUESTION 10

Students did not perform as well on this question as expected. Many students seemed to simply reorder the information from the question, suggesting that monkey A watches monkey B pick up the banana and the mirror neurons lead it to have the same activity as monkey B, with no evidence of any knowledge of their own being applied to the scenario.

Lots of students included an explanation of mirror neurons in isolation and whilst some of these showed good levels of detail, this was an application question and failing to refer to the scenario made these responses not creditworthy. Some ended up repeating the information when they went

on to explain mirror neurons in the context of the scenario costing them valuable time in the exam. Successful students presented their explanation of mirror neurons in the context of the monkey scenario from the start and continued to reference it throughout their answer.

Responses focused on learning and imitation, empathy and feeling, or imagination and memory, rather than mirror neuron activity, lacked effectiveness.

QUESTION 11

Over half of students produced high level 2 responses or above on this question, with just over a quarter of students achieving high level 3 or above. Defining theory of mind was almost always done accurately, and the connections to age and autism spectrum disorder (ASD) were included in most responses. In addition to this, the strongest responses featured detail knowledge of false-belief tasks including accurate findings for the different groups tested. This was followed by evaluation in the context of understanding autism, contradictory and complimentary research studies, and some methodological strengths and weaknesses of these studies always linked back to theory of mind and what it means for the explanation.

Quality of evaluation was consistently the element that held students back from achieving higher marks. It was often extremely generic in nature, or limited to methodological criticisms of studies not linked back to what this means for the explanation of social cognition.

A small number of students lost focus on the question and wrote extensively on ASD or mirror neurons which gained limited credit.

QUESTION 12

The majority of students achieved at least 1 mark on this question. The most selected incorrect option was as A followed by E.

QUESTION 13.1

This question was generally well answered. Again, some students included more information than required by this question explaining why pilot studies are conducted when they only needed to describe what they are.

QUESTION 13.2

When considered alongside some responses to question 13.1 it appears that understanding of this concept was weaker than expected for some of the students.

Whilst many clearly understood that the purpose of pilot studies was a procedural/methods trial to identify issues with the methodology and allow improvements, this was not always applied to the scenario as required by 'in this investigation' in the wording of the question. More effective responses talked about checking camera positioning or if the lecture was long enough to gather enough data.

However, many students presented responses suggesting that the same students would be used in a pilot study and real study or suggested that pilot studies allow for a hypothesis to be generated. A small number focused too much on the avoidance of unnecessary expense/wasting time rather than on what useful functions a pilot study performs.

QUESTION 14

This question was generally well answered, although there were a small number of students confusing sampling method in the context of participants, with sampling method in the context of observations (time vs. event), and some appear to lack knowledge of the names of recognised sampling methods in psychology.

Where students did not attain full marks, it tended to be due to a lack of explanation in the context of the scenario (application).

QUESTION 15.1

Students' explanations of this concept were often unclear meaning it was not always obvious that they understood the difference between a sample and the population. Those who were successful tended to present the explanation in terms of generalisation. When considered in the context of the success of students on question 15.2, it doesn't appear that there is an issue with understanding the concept, but rather the ability to explain it.

QUESTION 15.2

This question was answered correctly by most students.

QUESTION 16

This question was not well answered. There were lots of vague responses presenting the issues in very general terms or missing the focus of the question entirely. Many students simply defined the issues and/or offered solutions to them, rather than looking at them in the context of the scenario and identifying why the issues were present within this investigation.

Most commonly marks were achieved when focusing on embarrassment linked to sharing of test scores, although it wasn't always clear whether this was being presented as a confidentiality or protection from harm issue. Anxiety/embarrassment (caused by whatever aspect of the methodology) was credited as a protection from harm issue.

More successful students produced responses that identified a potential problem from the methodology in the scenario for each ethical issue, and explained what impact this issue would have on the participants, for example; linking video recordings to showing potentially private details or information and that sharing of this would be a failure to respect privacy, and that knowing they were being watched could provoke anxiety in participants which would be failing to protect them from harm.

QUESTION 17

Surprisingly, students struggled with this question, some even indicating on their papers that they did not understand it. Over a quarter of students either got 0 marks or did not attempt it.

Quite a few students referred to the test conducted after the lecture, but this was not part of the observation element of the scenario and is therefore not a suitable focus for this answer. Some simply repeated the category given in the question.

The clearest responses moved away from focusing on phone use and thought about other behaviours that could indicated attention (or lack of it) from students in this setting.

QUESTION 18

This question was generally well answered.

QUESTION 19.1

There were some excellent graphs presented by students with accurately named and scaled axes, and plotted points. The most dropped mark in these cases was for the title – ‘effect’ is not a synonym for correlation/relationship. Just under half of students achieved 3 marks or more on this question.

Some students produced bar graphs instead of scattergrams, but this was an error made by a minority. There were some strange scales used for the axes which then made plotting the points more difficult than it needed to be – more plotting errors were seen in these situations.

Students need to be reminded that when plotting points on a scattergram they need to do so clearly. Small dots or thin lines often did not differentiate plots on the axis from the axis/axis scale divisions, whilst oversized dots or crosses covering more than one small square couldn’t be deemed accurately plotted.

QUESTION 19.2

This question was generally well answered. However, some students simply described the graph and/or repeated data points they selected from the table rather than indicating what the graph was suggesting overall.

QUESTION 20.1

This question was generally well answered with over two thirds of students achieving full marks.

QUESTION 20.2

Students struggled with this question more than any other with only a third of the cohort achieving any marks and the largest number not attempting it when compared to other questions in this paper.

Many students were able to identify the issue of outliers but did not refer to the data from the table and/or go on to explain what impact an outlier would have on the range, especially in the context of the data set provided.

A misconception seen more than once was that a score of 0 means a range cannot be calculated. Other students focused on the small sample size, suggesting that makes the range unrepresentative, indicating a misunderstanding regarding the fact that the range is just representing the data set, not the wider population/frequency of behaviour. A small number explained how to calculate the range, lacking focus on the question asked.

QUESTION 21

References to demand characteristics and observer effects were seen in many responses, with the best responses identifying an issue and then explaining how it would affect the behaviour of the participants, or the data collected. Whilst students did not have to refer to the scenario in this question many who did were often successful.

Some students referred to experiments within their answer to this question. It is important that they are aware that observations and experiments are different research methodologies.